

ARRAY : Video - 171



CodeStoryWithMIK

Leetcode
- 1292

Medium

Something
important
you'll
learn



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Welcome to WeekendWithMIK

11iK.

They this channel to

see "Life behind the **scenes + Tech News**

Motivation -

The most powerful thing we have is that we have the ability to change our future in present by working hard. Failures in present are building blocks to a good future.



MIK...

1292. Maximum Side Length of a Square with Sum Less than or Equal to Threshold

Medium

Topics

Companies

Hint

Given a $m \times n$ matrix `mat` and an integer `threshold`, return the maximum side-length of a square with a sum less than or equal to `threshold` or return `0` if there is no such square.

Example:-

`mat` =

1	1	3	2	4	3	2
1	1	3	2	4	3	2
1	1	3	2	4	3	2

, threshold = 4

$2 \leq 4$

square side = 2

Output:

2

Constraints
Analysis:

Constraints:

- `m == mat.length` ✓ rows
- `n == mat[i].length` ✓ cols
- `1 <= m, n <= 300` → n^2, n^3
- `0 <= mat[i][j] <= 10^4`
- `0 <= threshold <= 10^5`

Thought Process

	0	1	2	3	4
0					
1					
2					
3					
4					
5					

rows = 6

cols = 5

$\leq$ threshold

(i+0, j+0)

off = 1

$\min(w, h)$

$w = 5 - j = 5 - 2 = 3$

$h = 6 - i = 6 - 2 = 4$

$\min(w, h) = 3$

offset = 0

Side = 0 + 1 = 1

bestSqSide = 0;

```

for(i = 0; i < m; i++) {
    for(j = 0; j < n; j++) {
        // (i, j)
    }
}

```

⇒ (i, j)

$O(m \cdot n \cdot \min(m, n) \cdot m \cdot n)$
 $O(m^2 n^2 \cdot \min(m, n))$
T.L.E.

```
for (int K = 0; K < min(rows-i, cols-j); K++) {
```

$r2 = i + K$; // opp. diag. cell.
 $c2 = j + K$;

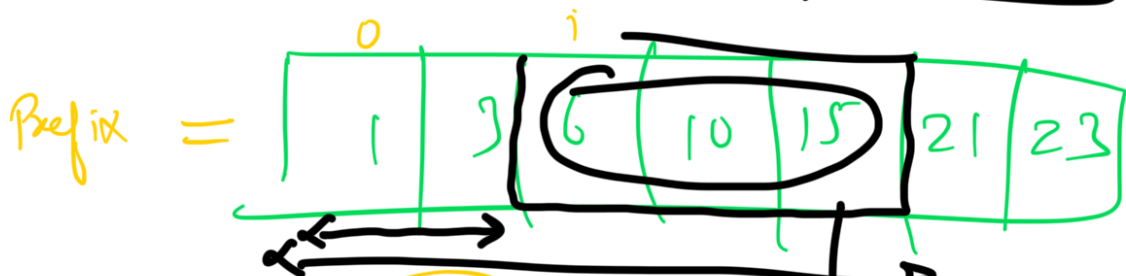
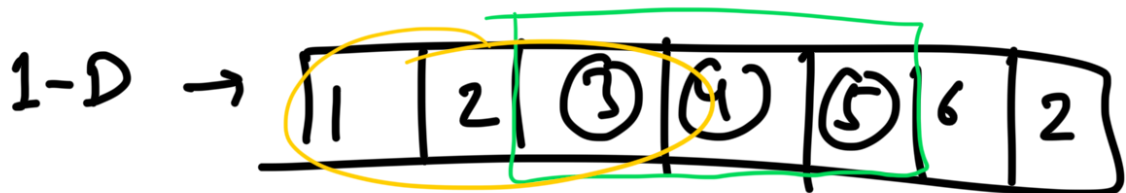
```
val = findSum(i, j, r2, c2);
```

```

if (val <= threshold) {
    bestSide = K + 1;
} else {
    break;
}

```

Optimally finding
 elements sum of
 a sub-matrix :-



Prefix[i]

$$15 - 3 = 12$$

1D Array $\rightarrow$ Sub-array $\rightarrow O(1)$

2D Array $\rightarrow$ Sub-matrix $\rightarrow O(1)$. $\leftarrow$

mat
prefix



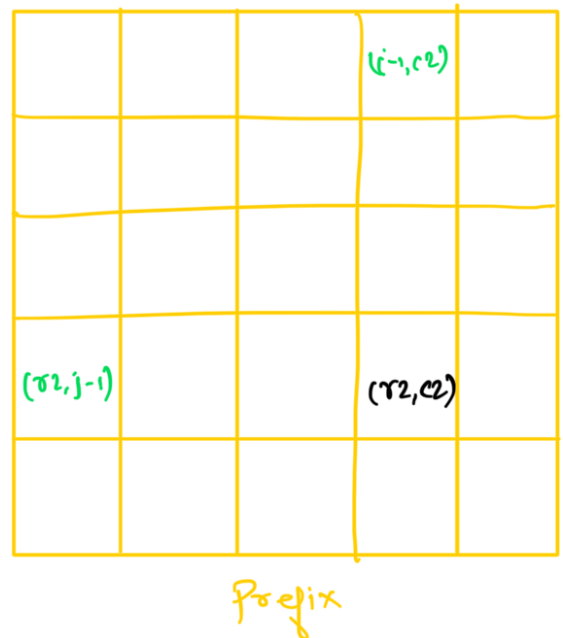
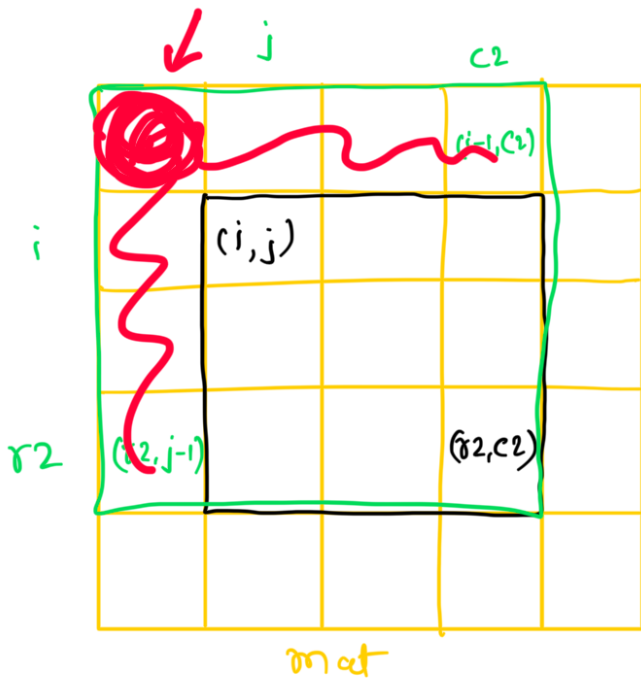
$$\begin{aligned} \text{Prefix}[i][j] &= \\ \text{Prefix}[i-1][j] &= \\ \text{Prefix}[i][j-1] &= \\ \text{Prefix}[i-1][j-1] &= \end{aligned}$$

$\text{Prefix}[0][0]$ = total sum of elements of mat from
top-left to (0,0)

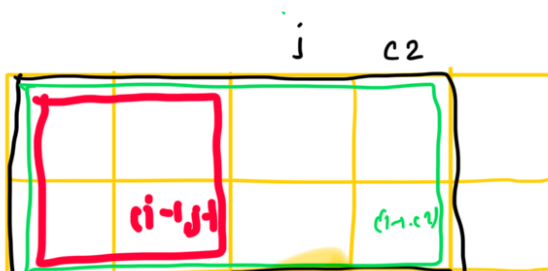
$\text{Prefix}[0][1]$ = total sum of elements of mat from
top-left to (0,1)

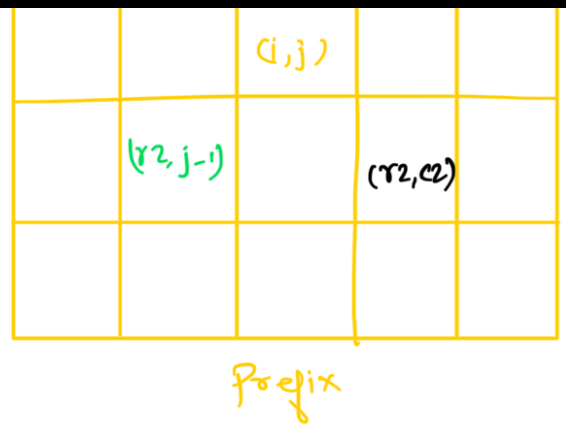
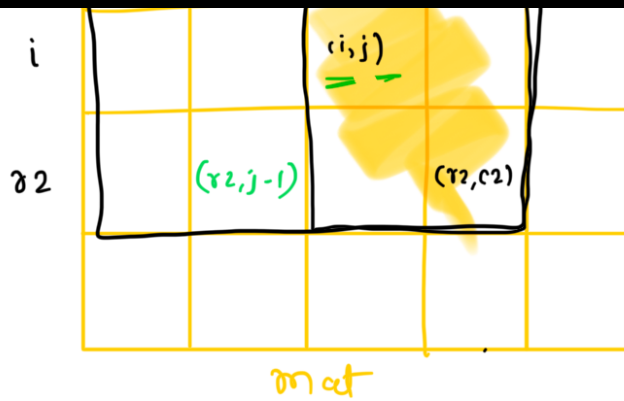
$\text{Prefix}[i][j]$ = total sum of elements of mat from
top-left to (i,j)
 $(0,0)$

$$\text{Prefix}[i][j] = \text{Prefix}[i-1][j] + \text{Prefix}[i][j-1] - \text{Prefix}[i-1][j-1] + \text{mat}[i][j];$$



$$\begin{aligned} &\text{Prefix}[r2][c2] \\ &- \text{Prefix}[i-1][c2] \\ &- \text{Prefix}[r2][j-1] \end{aligned}$$





Sum =

$$\begin{aligned} & \text{Prefix}[r2][c2] \\ & - \text{Prefix}[i-1][c2] \\ & - \text{Prefix}[r2][j-1] \\ & + \text{Prefix}[i-1][j-1] \end{aligned}$$

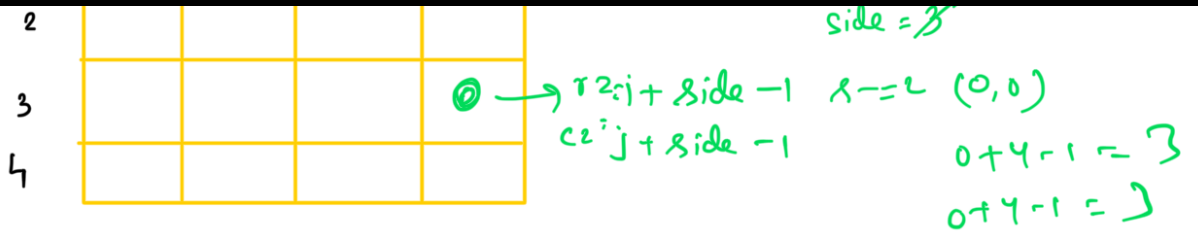
(sum <= the)

Approach - 2

	0	1	2	3
0				
1				

rows = 5
cols = 4

side = 4



$(i, j, r2, c2)$

Binary Search on side:

low = 1, high = min(row, col)

while (l <= h) {
 mid = l + (h - l) / 2;

if (check(mid)) {
 result = mid;
 low = mid + 1;
 }
 else
 high = mid - 1;

return result;

Constraints:

- `m == mat.length`
- `n == mat[i].length`
- `1 <= m, n <= 300`
- `0 <= mat[i][j] <= 104`
- `0 <= threshold <= 105`